

1. What is agile software development? Explain its key characteristics.

Agile software development is an approach to developing software through short and repeated development cycles instead of completing the whole project at once. The main idea is to deliver useful software regularly, receive feedback from customers, and make changes when requirements change. Agile focuses more on people, collaboration, working software, customer involvement, and responding to change than on following a fixed plan throughout the entire project (Beck et al., 2001).

Some important characteristics of Agile software development are:

- Iterative development: Software is developed in small iterations, allowing teams to improve the product gradually.
- Customer collaboration: Customers or stakeholders are involved throughout development so their feedback can influence the product.
- Flexibility: Agile allows requirements to change even after development has started.
- Frequent delivery: Working software is delivered regularly instead of waiting until the end of the project.
- Team collaboration: Developers, customers, and other stakeholders communicate frequently and work together.
- Continuous improvement: Teams regularly review their work and identify ways to improve their processes and product.

Agile can be useful when requirements aren't completely clear at the beginning of a project. However, it requires good communication and active participation from team members and customers (Beck et al., 2001).

2. Why do we need Extreme Programming? Explain its benefits and challenges.

Extreme Programming (XP) is an Agile software development method that focuses strongly on software quality, customer satisfaction, frequent releases, and continuous feedback. XP was developed to help software teams respond effectively when requirements change frequently. It uses practices such as pair programming, test-driven development, continuous integration, simple design, and frequent customer feedback (Beck, 2000).

XP is useful because software requirements can change during development. Instead of spending a long time creating a complete system based on assumptions, an XP team develops smaller features, tests them, receives feedback, and improves them. This can reduce the risk of developing features that don't meet customer needs.

The main benefits of XP include:

- Better software quality: Frequent testing and continuous integration can help identify problems earlier.
- Quick feedback: Customers can provide feedback regularly.
- Adaptability: Teams can respond more easily to changing requirements.
- Reduced defects: Practices such as test-driven development and pair programming can help detect errors earlier.
- Continuous delivery: Small releases allow useful features to reach users sooner.

However, XP also has some challenges. It requires strong communication and active customer involvement, which may not always be available.

Pair programming can also require more developer time because two developers work together on the same task. In addition, practices such as continuous testing and frequent integration require discipline from the development team. Therefore, XP may be difficult to apply effectively when the team or organization isn't prepared to follow its practices consistently (Beck, 2000).

3. What is Scrum in Agile? Explain its key roles (Scrum Master, Product Owner, Development Team).

Scrum is an Agile framework used to help teams develop and deliver products through short, controlled periods of work called Sprints. A

Sprint is a fixed period during which the team works toward a specific goal and produces a usable product increment. Scrum emphasizes transparency, inspection, and adaptation, allowing teams to regularly examine their progress and make improvements (Schwaber & Sutherland, 2020).

The main Scrum roles are:

Role	Main Responsibility
Scrum Master	Helps the team understand and follow Scrum, removes obstacles, and supports effective teamwork.
Product Owner	Represents product interests, manages and orders the Product Backlog, and helps maximize product value.

Developers	Build the product increment and are responsible for creating a usable product that meets the Sprint Goal.
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The Scrum Master doesn't act as a traditional manager. Instead, this role helps the Scrum Team work effectively and supports the removal of barriers that could prevent progress. The Product Owner is responsible for maximizing the value of the product and managing the Product Backlog. The Developers are the people who create the product increment during the Sprint (Schwaber & Sutherland, 2020).

Overall, Agile provides a flexible approach to software development, while XP provides specific engineering and development practices, and Scrum provides a framework for organizing Agile product development. These approaches can help teams respond to changing requirements, improve communication, and deliver useful software more frequently.

References

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